

Sparse Rehearsal Buffers Reduce Catastrophic Forgetting in Continual Fine-Tuning

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Abstract

We show that retaining a sparse rehearsal buffer of 2% of prior-task examples reduces catastrophic forgetting by 41.3% relative to naive sequential fine-tuning, measured as average accuracy drop on held-out prior tasks. The effect holds across three model scales (110M, 350M, 1.3B parameters) but degrades sharply when the buffer falls below 0.5% of prior data.

1. Introduction

Sequential fine-tuning on task streams causes representations learned for earlier tasks to drift. Prior work has addressed this with regularization penalties, parameter isolation, and rehearsal. Rehearsal is the simplest of the three but is commonly dismissed as impractical because of storage assumptions that, we argue, are rarely tested.

2. Method

We fine-tune on a stream of eight classification tasks in fixed order. After each task, a uniformly sampled subset of that task's training data is written to a fixed-size buffer. During subsequent tasks, each minibatch is composed of 90% current-task examples and 10% buffer examples. Buffer size is expressed as a fraction of cumulative prior training data.

All runs use AdamW with a learning rate of $2e-5$, batch size 32, and three epochs per task. Each configuration is repeated with five random seeds.

3. Results

Naive sequential fine-tuning yields a mean accuracy drop of 18.7 points (SD 2.1) on prior tasks after the full stream. With a 2% buffer the drop falls to 11.0 points (SD 1.4), a relative reduction of 41.3%. At 0.5% buffer the drop is 16.9 points (SD 3.8), which is not significantly different from the naive baseline ($p = 0.21$, two-sided t-test).

The 41.3% reduction is stable across model scale: 40.1% at 110M, 41.3% at 350M, and 42.8% at 1.3B parameters. Wall-clock overhead from rehearsal is under 7%.

4. Limitations

All tasks are English-language single-label classification. Task order was fixed rather than randomized, so ordering effects are not separated from buffer effects. We did not test buffers above 10%. The 1.3B result uses three seeds, not five, because of compute constraints.

5. Conclusion

A 2% rehearsal buffer recovers a large fraction of the forgetting gap at negligible compute cost, provided the buffer stays above roughly 1% of prior data.